Branch Cash Flow Insights

Business Problem Statement

- The bank is facing inefficiencies in cash flow management across branches, leading to cash shortages and uneven cash distribution.
- Certain branches show consistently high withdrawal volumes, indicating potential liquidity pressure and risk of stockouts.
- There is a mismatch between deposits and withdrawals , but the net cash flow is not optimally distributed across regions and branches.
- Monthly trends reveal fluctuations in deposits, making it difficult to accurately forecast cash requirements.
- Lack of proactive monitoring leads to operational inefficiencies, excess idle cash in some branches, and shortages in others.

Project Overview

Branch Cash Flow Insights focuses on understanding how cash inflows, outflows, and branch-level liquidity patterns impact operational efficiency and customer service in banking systems. The project identifies key indicators of inefficiency, such as recurring cash shortages, overstocked branches, and unusual cash movement trends.

By analyzing daily deposits, withdrawals, inter-branch transfers, and ATM transactions, it uncovers patterns of cash utilization, branch performance disparities, and potential operational risks. These insights help optimize cash allocation, reduce idle cash, prevent shortages, and enhance overall efficiency in branch operations.

Dataset Summary

- Rows: ~3,000+
- Columns: 10+
- Branch details: Branch ID, Region
- Cash transaction data: Deposit Amount, Withdrawal Amount
- Balance: Opening Balance, Closing Balance
- Customer activity metrics: Transaction Count
- Operational indicators: Cash Shortage Flag

Exploratory Data Analysis using Python

- Data Cleaning using pandas**:** Handled null values, standardized cash amounts, removed duplicates, and formatted date columns for accurate time-series analysis.

- Feature Engineering: Created branch-level features such as Cash Difference (Deposits vs Withdrawals), Net Cash Flow, and extracted date features (day, week, month).

- Exploratory Analysis: Analyzed cash inflows/outflows, branch-wise cash patterns, ATM vs counter transactions, peak hours for withdrawals, and inter-branch transfer trends using seaborn and matplotlib.
- Anomaly Detection: Identified branches with recurring cash shortage, and operational inefficiencies.

- SQL Integration: Loaded the cleaned and processed data into a SQL database for further analysis.

Data Analysis using SQL (Business Questions)

• Total Cash Deposits by Region

	Region text	sum double precision
1	South	202136870.5
2	North	214480768.5
3	West	214451191
4	East	207258254

• Branches with Highest Withdrawal – Identified Top 5 Branches with Max Withdrawal

	Branch_ID text	total_withdrawal double precision
1	B29	18662215
2	B35	18442795
3	B8	18391681
4	B32	18292108
5	B26	17541450

• Count of Cash Shortage – Identified how many Branches have Cash Shortage

	count bigint
1	129

• Average Balance by Region – Identified Avg Balance per Region

	Region text	avg double precision
1	South	540529.078581363
2	North	533618.7554557124
3	West	540248.3326666667
4	East	526599.1043882979

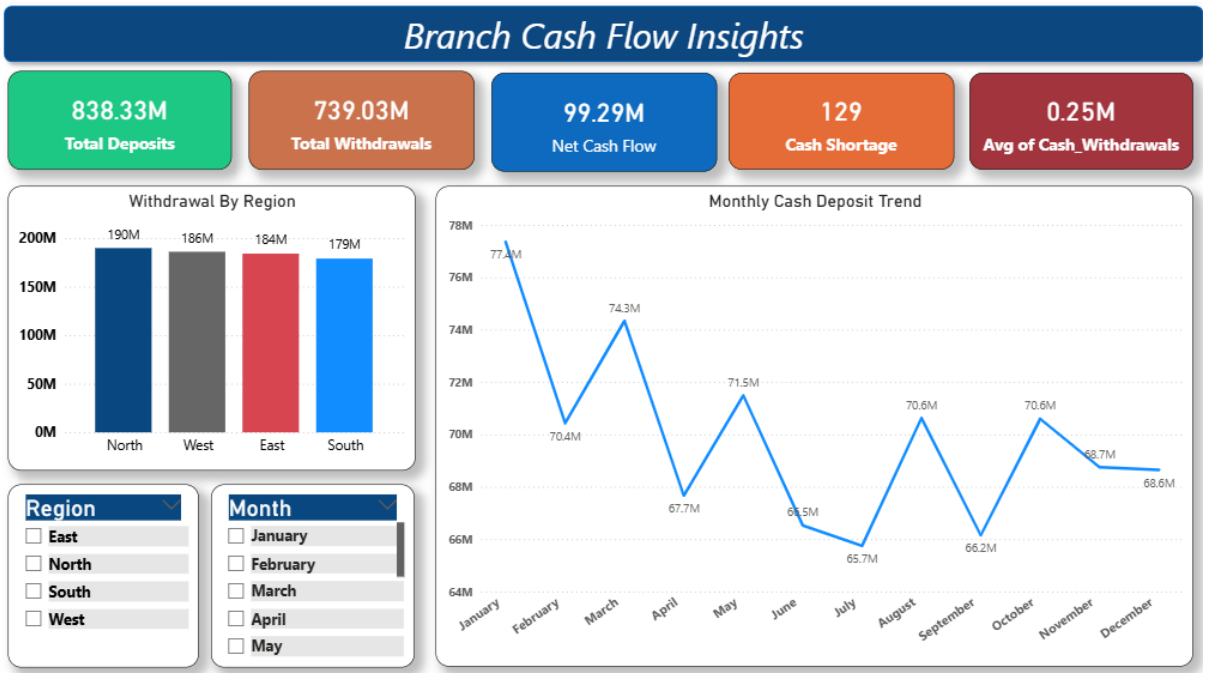
• Monthly Total Withdrawal Trend – Identified Monthly Total Withdrawal Trend as per Branch

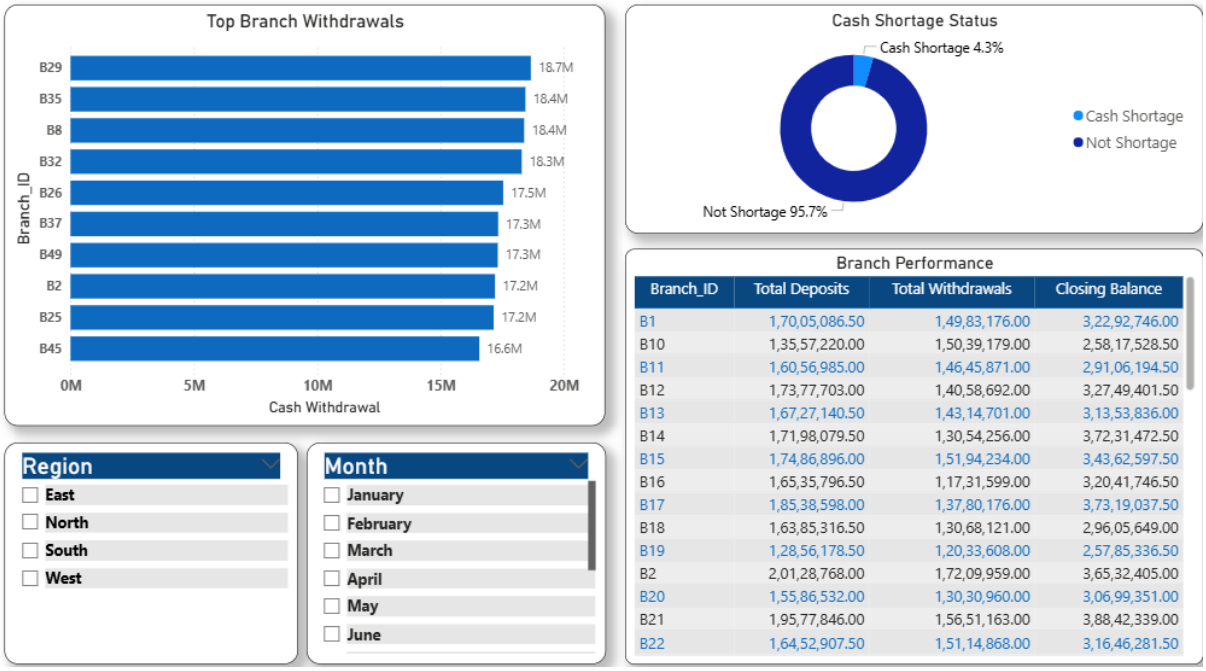
	month_name text	total_withdrawals double precision
1	January	67004612
2	February	63483335
3	March	69801916
4	April	58059555
5	May	58624308
6	June	61782443

	month_name text	total_withdrawals double precision
7	July	60932841
8	August	62924176
9	September	56008178
10	October	63509849
11	November	55750161
12	December	61151402

Dashboard in Power BI

Developed an interactive Power BI dashboard to monitor cash flow KPIs, analyze branch-wise deposits and withdrawals, identify high-risk branches with cash shortages, track regional cash distribution, and visualize monthly cash trends to enable efficient liquidity management and data-driven decision-making.





Branch_ID	Total Deposits	Total Withdrawals	Closing Balance
B1	1,70,05,086.50	1,49,83,176.00	3,22,92,746.00
B10	1,35,57,220.00	1,50,39,179.00	2,58,17,528.50
B11	1,60,56,985.00	1,46,45,871.00	2,91,06,194.50
B12	1,73,77,703.00	1,40,58,692.00	3,27,49,401.50
B13	1,67,27,140.50	1,43,14,701.00	3,13,53,836.00
B14	1,71,98,079.50	1,30,54,256.00	3,72,31,472.50
B15	1,74,86,896.00	1,51,94,234.00	3,43,62,597.50
B16	1,65,35,796.50	1,17,31,599.00	3,20,41,746.50
B17	1,85,38,598.00	1,37,80,176.00	3,73,19,037.50
B18	1,63,85,316.50	1,30,68,121.00	2,96,05,649.00
B19	1,28,56,178.50	1,20,33,608.00	2,57,85,336.50
B2	2,01,28,768.00	1,72,09,959.00	3,65,32,405.00
B20	1,55,86,532.00	1,30,30,960.00	3,06,99,351.00
B21	1,95,77,846.00	1,56,51,163.00	3,88,42,339.00
B22	1,64,52,907.50	1,51,14,868.00	3,16,46,281.50

Key Insights from Dashboard

- North region has the highest withdrawals (~190M) → high demand zone
- Top branches (B29, B35, B8) show peak withdrawals → high-risk for shortages
- Cash shortage rate: 4.3% → indicates liquidity gaps
- Deposit trend fluctuates (65M–77M range) → inconsistent inflow patterns
- Some branches maintain high closing balances → indicates idle cash

Business Recommendation

- Reduced cash shortages and customer dissatisfaction
- Improved liquidity planning and operational efficiency
- Lower idle cash holding costs
- Better data-driven decision-making for branch operations
- Regional Strategy
- Focus on High-Risk Branches
- Optimize Cash Allocation
- Improve Cash Forecasting